ECG Arrhythmia Classification Using Advanced Temporal Pattern Recognition

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Biomedical Signal Processing | Machine Learning | Graphical Abstract

PROBLEM STATEMENT

An **electrocardiogram (ECG)** records the heart's electrical activity as a repeating **P – QRS – T** waveform; each R-peak marks one heartbeat.

A **cardiac arrhythmia** is a deviation from normal sinus rhythm – from a benign variant to a marker of a serious underlying condition.

Why it matters: telling clinically important abnormal beats from normal ones, reliably and at scale, underlies Holter analysis and wearable monitoring.

Why it is hard (three compounding challenges):

- **Class imbalance** – normal beats are ~89% of the data, so a useless classifier can still look accurate.
- **Inter-patient variability** – the same arrhythmia looks different across people; if a patient's beats leak into training, accuracy is inflated.
- **Temporal dependency** – premature beats are defined by *timing* relative to neighbours, others by waveform *shape*.

PROJECT MOTIVATION

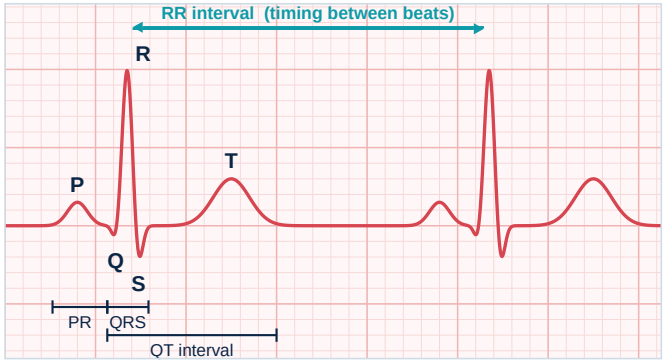
Scalable cardiac monitoring needs **automated, data-driven beat classification that is honestly evaluated**. This project compares three families of approach – **beat timing** (RR/HRV), **frequency-domain morphology** (wavelets) and **learned raw-waveform representations** (CNN + BiLSTM) – under one leak-free protocol, so differences reflect the model, not the data handling.

PROJECT OBJECTIVES

1. **Implement & compare** RR/HRV + SVM, DWT + Random Forest and 1D-CNN + BiLSTM on *identical* preprocessed beats.
2. **Patient-independent evaluation** (de Chazal DS1/DS2): no patient in both sets; all tuning on DS1 only.
3. **Imbalance-aware reporting**: macro-F1 and per-class precision/recall, not accuracy alone.
4. **Explain disagreements** between approaches using the actual confusion matrices.

ECG WAVEFORM ANATOMY

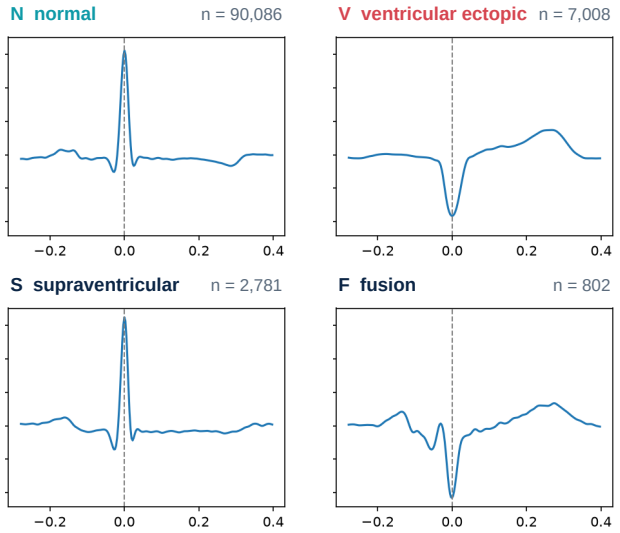
idealised schematic



Grid: 1 small square = 0.04 s / 0.1 mV. P = atrial depolarisation; QRS = ventricular depolarisation; T = repolarisation. Schematic only – not project data.

NORMAL VS. ABNORMAL BEATS

real project data



Segmented, z-score-normalised MLII beats (shared y-scale, x = seconds relative to the R-peak) from the project's own processed dataset. V beats are wide with a different morphology; S beats resemble N. Q (n = 15) omitted.

DATASET OVERVIEW

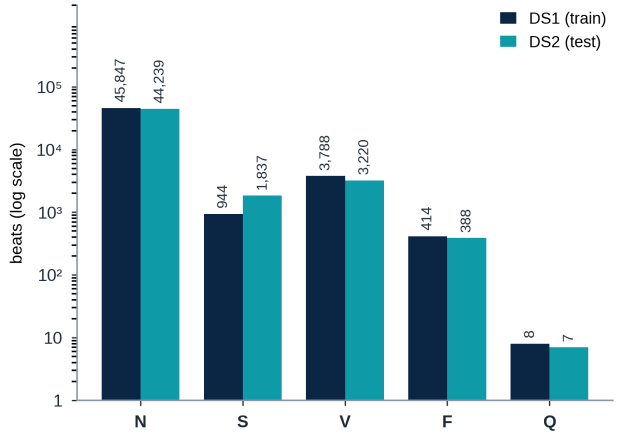
MIT-BIH Arrhythmia Database

Source	PhysioNet – Beth Israel Hospital Arrhythmia Laboratory (Moody & Mark, 2001; Goldberger et al., 2000)
Recordings	48 half-hour, two-lead ambulatory ECGs (47 subjects); **44 used** – paced-dominated 102, 104, 107, 217 excluded
Signal	360 Hz, 11-bit, ±5 mV; **lead MLII**; cardiologist beat annotations
Samples	**100,692 beats** → fixed 244-sample windows
Labels	AAMI EC57 **5 classes**: N, S, V, F, Q
Split	de Chazal inter-patient: **DS1** 22 records (train) / **DS2** 22 records (test)

DS1 train · 51,001 beats | **DS2 test · 49,691 beats**

CLASS DISTRIBUTION

AAMI EC57 classes



N normal / bundle-branch block · **S** supraventricular ectopic · **V** ventricular ectopic · **F** fusion · **Q** unknown / paced.

N is 89% of DS2; Q has single-digit support (8 / 7 beats).

Proposed Methodology & Technical Architecture

From raw MIT-BIH signal to five-class AAMI prediction

SHARED PREPROCESSING PIPELINE

identical for all three approaches – only stages implemented in src/ are shown

1 Acquire & load

PhysioNet via **wfdb**: 44 records, MLII lead, 360 Hz; exploratory analysis in a notebook.

Why: *Cardiologist-annotated, reproducible source.*

2 Baseline-wander removal

Two-stage **median filter** (≈200 ms, then ≈600 ms); estimate is subtracted.

Why: *Removes slow drift without distorting QRS / ST.*

3 Band-pass filtering

3rd-order **Butterworth 0.5–40 Hz**, zero-phase (filtfilt).

Why: *Cuts muscle noise & 60 Hz mains; no phase distortion.*

4 Beat segmentation

0.28 s before + 0.40 s after each annotated R-peak = **244 samples**; invalid windows dropped; symbols → AAMI class.

Why: *Fixed-length input covering QRS and T-wave.*

5 Normalise & RR timing

Per-beat **z-score**; previous / next **RR interval** from beat annotations.

Why: *Removes gain differences; supplies timing.*

6 Patient-independent split

DS1 (22 rec.) train / **DS2** (22 rec.) test; tuning by **GroupKFold** within DS1.

Why: *Prevents patient leakage.*

Same cleaned beats + same DS1 / DS2 split feed three parallel approaches – they differ only in the representation given to the classifier

APPROACH 1 · RR / HRV + SVM

timing features

6 RR/HRV features (per beat) → **StandardScaler** (z-score) → **SVC · RBF kernel** (class_weight = balanced) → **N S V F Q** prediction

Features: RR_{prev}, RR_{next}, RR_{mean}, RR_{ratio}, $|RR_{next} - RR_{prev}|$ (local RMSSD proxy), instantaneous heart rate.
Tuning: C in {1, 10, 100}, γ in {scale, 0.01, 0.1}; 5-fold GroupKFold within DS1, scored by macro-F1. **Trained on** a class-balanced DS1 subsample (≤ 4,000 beats/class = 9,154 beats).

APPROACH 2 · DWT + RANDOM FOREST

morphology features

db4 DWT, level 4 (5 coefficient bands) → **30 statistics** (5 bands × 6 stats) → **Random Forest** (class_weight = balanced) → **N S V F Q** prediction

Features: bands cA_4, cD_4–cD_1 of the normalised beat; per band mean, std, min, max, energy and Shannon entropy.
Tuning: n_estimators in {200, 400}, max_depth in {None, 20}, min_samples_leaf in {1, 3}; same GroupKFold search. **Trained on** the full DS1 set (51,001 beats).

APPROACH 3 · 1D-CNN + BIDIRECTIONAL LSTM (TEMPORAL PATTERN LEARNING, END-TO-END ON THE RAW BEAT)

PyTorch · src/model_cnn_bilstm.py

Input — normalised raw beat — `(1 × 244)`

→ **Conv block 1** — Conv1D 16, k = 7; ReLU · MaxPool 2 — `(16 × 122)`

→ **Conv block 2** — Conv1D 32, k = 5; ReLU · MaxPool 2 — `(32 × 61)`

→ **BiLSTM** — hidden 64 per direction; batch-first — `(61 × 128)`

→ **Last step + Dropout** — final time-step output; p = 0.3 — `(128)`

→ **Dense + Softmax** — Linear 128 → 5 — **N · S · V · F · Q**

Shapes are (channels × length) from the implemented model; the LSTM reads the 61-step feature sequence in both directions.

TEMPORAL MECHANISM

Convolution filters learn local waveform morphology; the BiLSTM models how the feature sequence evolves across the beat in forward and backward directions.

TRAINING

Adam (lr = 1e-3), batch 128, ≤ 25 epochs; early stopping (patience 5) on **validation macro-F1**; cross-entropy with class weights $N/(5 \cdot n_c)$ capped at 10×.

DATA & COMPUTE

DS1 split 85 % / 15 % **by record** (GroupShuffleSplit) for train / validation; DS2 only for final test; fixed seed 42; run on CPU.

EVALUATION & INFERENCE

DS2 touched once

Test set: DS2, 49,691 beats from 22 unseen patients, evaluated once after all tuning on DS1.

Reported: accuracy, macro- & weighted-F1, per-class precision / recall / F1, confusion matrices, one-vs-rest macro ROC-AUC. Specificity is not reported.

Prediction: `predict.py` and a Streamlit demo segment a MIT-BIH record and output a class and confidence per beat (offline demo).

TECHNOLOGY STACK

requirements.txt

Python 3.12 · wfdb · NumPy · pandas · SciPy · PyWavelets · scikit-learn · PyTorch · joblib · Matplotlib · Seaborn · Streamlit

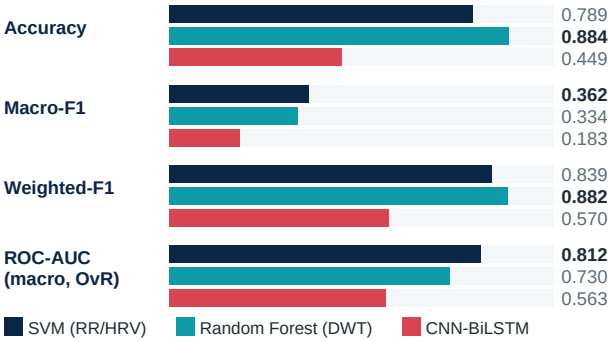
language · data & arrays · signal processing · ML / deep learning · utilities & plots · demo UI

Experimental Results, Findings & Conclusion

DS2 test set · 49,691 beats · 22 patients unseen during training

AGGREGATE PERFORMANCE

DS2



Bold = best per metric. Bars start at 0.

CONFUSION MATRICES (DS2)

counts shown · shade = row-normalised (recall)

SVM (RR/HRV) acc 0.789

true \ predicted	N	S	V	F	Q
N	**36,028**	546	953	6,178	534
S	1,119	**84**	321	50	263
V	88	126	**2,739**	250	17
F	15	4	3	**366**	0
Q	5	0	1	0	**1**

Random Forest (DWT) acc 0.884

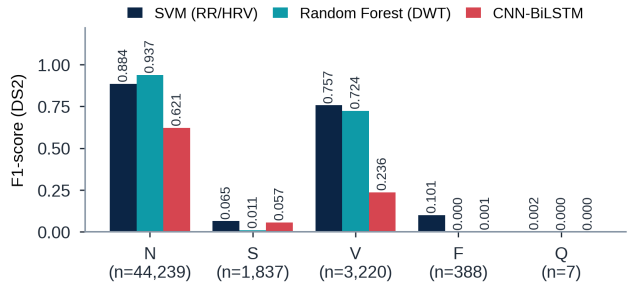
true \ predicted	N	S	V	F	Q
N	**41,233**	136	1,316	1,551	3
S	1,798	**11**	19	9	0
V	523	1	**2,690**	6	0
F	205	0	183	**0**	0
Q	1	0	6	0	**0**

CNN-BiLSTM acc 0.449

true \ predicted	N	S	V	F	Q
N	**20,335**	5,439	8,274	10,191	0
S	196	**233**	1,338	70	0
V	640	445	**1,737**	398	0
F	76	154	154	**4**	0
Q	3	3	1	0	**0**

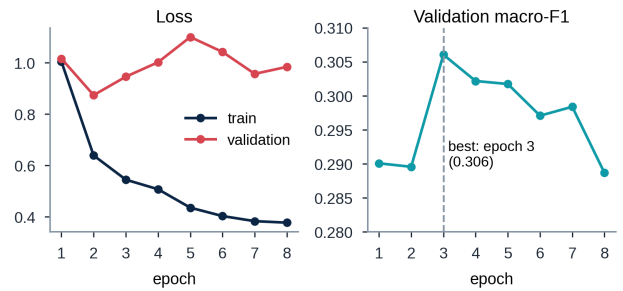
PER-CLASS F1-SCORE

DS2



CNN-BILSTM TRAINING CURVES

DS1 train / val



KEY FINDINGS

1. **No single winner:** RF leads accuracy (0.884) and weighted-F1 (0.882); SVM leads macro-F1 (0.362) and ROC-AUC (0.812); CNN-BiLSTM trails (0.449 / 0.183).
2. **Accuracy misleads:** class N is 89% of DS2, so macro-F1 and per-class recall decide the ranking.
3. **V is the best-detected abnormal class** for the classical models (F1 0.757 SVM, 0.724 RF); CNN-BiLSTM reaches 0.236.
4. **RR timing is very sensitive to F:** SVM recall 0.943 but precision 0.053 (6,178 N beats flagged as F).
5. **S is hard for all models** (F1 ≤ 0.065); Q has 7 test beats and is not statistically reliable.

LIMITATIONS

- **Q** has only 8 (DS1) / 7 (DS2) beats once paced-dominated records are excluded.
- **SVM** trained on a class-balanced DS1 subsample (≤ 4,000 beats/class; 9,154 beats).
- **CNN-BiLSTM** is deliberately compact and briefly trained (early-stopped at epoch 8 of ≤ 25; best-validation checkpoint from epoch 3 used; class weights capped at 10×).
- **Single lead (MLII)** from one 1970s database of 47 subjects; no external validation.
- **Offline, retrospective** prototype – not a clinically validated tool.

FUTURE SCOPE

- Feed an **explicit RR-interval channel** and/or **multi-lead** input to the CNN-BiLSTM.
- **Larger, longer-trained** deep model with a learning-rate schedule.
- **Attention-based** temporal models (e.g. a small Transformer encoder).
- **External validation** on a second database (e.g. INCART, PTB-XL).
- Imbalance-aware losses and **explainability** (attention maps, SHAP).

CONCLUSION

Under one leak-free, patient-independent protocol, RR/HRV + SVM achieved the best macro-F1 and ROC-AUC, DWT + Random Forest led on accuracy by excelling at class N, and the compact CNN-BiLSTM trailed both. The central takeaway is methodological: under severe imbalance, accuracy alone can hide near-zero recall on minority classes – only per-class, inter-patient evaluation reveals it. All values come from executed experiments.

References (from the project's references.bib) · [1] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH Arrhythmia Database," *IEEE Eng. Med. Biol. Mag.*, 20(3):45–50, 2001. [2] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet," *Circulation*, 101(23):e215–e220, 2000; dataset: physionet.org/content/mitdb/1.0.0/. [3] P. de Chazal, M. O'Dwyer, R. B. Reilly, "Automatic classification of heartbeats using ECG morphology and heartbeat interval features," *IEEE Trans. Biomed. Eng.*, 51(7):1196–1206, 2004. [4] AAMI, ANSI/AAMI EC57, 2012. [5] E. Alickovic and A. Subasi, "Medical decision support system for diagnosis of heart arrhythmia using DWT and random forests classifier," *J. Med. Syst.*, 40:108, 2016. [6] U. R. Acharya et al., "A deep convolutional neural network model to classify heartbeats," *Comput. Biol. Med.*, 89:389–396, 2017. [7] F. Liu et al., "An attention-based hybrid LSTM-CNN model for arrhythmias classification," *IJCNN*, 2019. [8] M. G. Tsipouras et al., *Artif. Intell. Med.*, 33(3):237–250, 2005. [9] N. Kohli et al., *ICCCT*, pp. 486–490, 2010.